Table 1: RN masks were reconstructed from exact archived data/code because selected indices were not originally stored. CatBoost results use the final train+validation RN selection stage. RN purity is true negatives / selected U; true-negative recall is selected true negatives / available true negatives (also the fraction of available true negatives retained). U retention is selected U / available U and is shown separately. The figure averages settings within each of ten families; the table retains all 40 settings per design. Interaction is explicitly labelled. AUROC/AP changes are RN CatBoost minus naive CatBoost; no broad BA comparison is made.

One-sample Scenario	RN purity	TN retained / available	U retention	AUROC change (pp)	AP change (pp)	n
Reference	0.964	0.212	0.188	+0.23	-1.02	100
Prevalence $\pi = 0.01$	0.997	0.354	0.352	+4.09	+0.42	100
Prevalence $\pi = 0.05$	0.992	0.209	0.204	+1.20	-0.58	100
Prevalence $\pi = 0.10$	0.983	0.208	0.197	+0.40	-1.44	100
Prevalence $\pi = 0.15$	0.974	0.212	0.194	+0.29	-1.45	100
Prevalence $\pi = 0.25$	0.952	0.211	0.179	+0.17	-1.05	100
Prevalence $\pi = 0.30$	0.939	0.212	0.173	+0.11	-0.77	100
Prevalence $\pi = 0.35$	0.925	0.207	0.163	+0.13	-0.67	100
Prevalence $\pi = 0.40$	0.908	0.208	0.157	+0.10	-0.49	100
Prevalence $\pi = 0.45$	0.892	0.206	0.147	+0.11	-0.37	100
Prevalence $\pi = 0.50$	0.868	0.210	0.143	+0.19	-0.20	100
Labeling probability $c = 0.05$	0.942	0.227	0.195	+2.47	+1.48	100
Labeling probability $c = 0.10$	0.953	0.195	0.168	+1.21	-0.30	100
Labeling probability $c = 0.15$	0.955	0.207	0.179	+0.61	-0.92	100
Labeling probability $c = 0.20$	0.958	0.204	0.178	+0.38	-1.45	100
Labeling probability $c = 0.25$	0.961	0.211	0.185	+0.33	-1.07	100
Labeling probability $c = 0.35$	0.966	0.210	0.187	+0.11	-1.19	100
Labeling probability $c = 0.40$	0.968	0.217	0.195	+0.03	-1.30	100
Labeling probability $c = 0.45$	0.970	0.221	0.200	+0.04	-1.18	100
Labeling probability $c = 0.50$	0.974	0.213	0.194	-0.03	-1.11	100
Class separation $\delta = 0.50$	0.935	0.107	0.097	+0.41	-0.42	100
Class separation $\delta = 0.70$	0.952	0.150	0.134	+0.25	-0.86	100
Class separation $\delta = 1.20$	0.974	0.310	0.271	+0.19	-1.52	100
High dimension ($p = 50$)	0.962	0.178	0.158	+0.54	-0.34	100
High dimension ($p = 100$)	0.961	0.146	0.130	+0.59	-0.27	100
High dimension ($p = 500$)	0.942	0.080	0.072	+0.60	-0.14	100
Correlation $\rho = 0.30$	0.964	0.206	0.182	-0.21	-2.44	100
Correlation $\rho = 0.60$	0.965	0.208	0.184	+0.31	-1.46	100
Prevalence $\pi = 0.01$, $c = 0.50$	0.998	0.295	0.294	+3.38	+0.11	100
Prevalence $\pi = 0.05$, $c = 0.10$	0.986	0.293	0.284	+3.55	+1.35	100
Prevalence $\pi = 0.05$, $c = 0.50$	0.994	0.215	0.211	+0.63	-0.91	100
Prevalence $\pi = 0.10$, $c = 0.05$	0.970	0.256	0.239	+4.11	+2.63	100
Prevalence $\pi = 0.10$, $c = 0.10$	0.975	0.234	0.219	+2.02	+0.35	100
Prevalence $\pi = 0.10$, $c = 0.50$	0.988	0.213	0.205	+0.13	-1.65	100
Prevalence $\pi = 0.50$, $c = 0.05$	0.830	0.153	0.096	+1.99	+1.07	100
Prevalence $\pi = 0.50$, $c = 0.10$	0.838	0.178	0.113	+0.88	+0.12	100
Prevalence $\pi = 0.50$, $c = 0.50$	0.905	0.214	0.157	-0.09	-0.46	100
Curved boundary	0.957	0.172	0.153	-1.06	-2.86	100
Feature interaction Interaction	0.944	0.135	0.122	-10.00	-13.16	100
Unequal class covariance	0.938	0.129	0.117	-4.34	-9.24	100

Two-sample Scenario	RN purity	TN retained / available	U retention	AUROC change (pp)	AP change (pp)	n
Reference	0.950	0.205	0.174	+0.21	-1.14	100
Prevalence $\pi = 0.01$	0.998	0.210	0.208	+0.01	-0.38	100
Prevalence $\pi = 0.05$	0.988	0.217	0.209	+0.02	-1.10	100
Prevalence $\pi = 0.10$	0.976	0.217	0.201	+0.14	-1.31	100
Prevalence $\pi = 0.15$	0.963	0.213	0.188	+0.24	-0.99	100
Prevalence $\pi = 0.25$	0.934	0.198	0.160	+0.17	-1.19	100
Prevalence $\pi = 0.30$	0.914	0.203	0.156	+0.25	-0.93	100
Prevalence $\pi = 0.35$	0.897	0.197	0.144	+0.30	-0.90	100
Prevalence $\pi = 0.40$	0.875	0.194	0.134	+0.39	-0.62	100
Prevalence $\pi = 0.45$	0.850	0.190	0.124	+0.58	-0.35	100
Prevalence $\pi = 0.50$	0.826	0.173	0.106	+0.68	-0.04	100
Labeled positives $n_P = 50$	0.942	0.206	0.176	+2.76	+1.93	100
Labeled positives $n_P = 100$	0.948	0.189	0.160	+1.11	-0.49	100
Labeled positives $n_P = 150$	0.947	0.205	0.174	+0.69	-0.97	100
Labeled positives $n_P = 200$	0.947	0.209	0.177	+0.49	-0.88	100
Labeled positives $n_P = 250$	0.948	0.209	0.176	+0.32	-1.15	100
Labeled positives $n_P = 350$	0.950	0.198	0.167	+0.19	-1.18	100
Labeled positives $n_P = 400$	0.952	0.198	0.166	+0.07	-1.13	100
Labeled positives $n_P = 450$	0.949	0.208	0.176	+0.10	-0.94	100
Labeled positives $n_P = 500$	0.948	0.217	0.183	+0.09	-0.98	100
Class separation $\delta = 0.50$	0.913	0.096	0.084	+0.43	-0.33	100
Class separation $\delta = 0.70$	0.932	0.154	0.132	+0.33	-0.73	100
Class separation $\delta = 1.20$	0.964	0.306	0.254	+0.20	-1.17	100
High dimension ($p = 50$)	0.946	0.172	0.146	+0.63	-0.02	100
High dimension ($p = 100$)	0.944	0.144	0.122	+0.71	-0.03	100
High dimension ($p = 500$)	0.916	0.083	0.073	+0.75	+0.16	100
Correlation $\rho = 0.30$	0.948	0.208	0.176	-0.14	-2.08	100
Correlation $\rho = 0.60$	0.949	0.218	0.184	+0.42	-0.94	100
Prevalence $\pi = 0.01$, $n_P = 25$	0.997	0.281	0.279	+3.38	+0.21	100
Prevalence $\pi = 0.01$, $n_P = 100$	0.998	0.204	0.202	+0.87	-0.45	100
Prevalence $\pi = 0.01$, $n_P = 500$	0.998	0.221	0.219	-0.16	-0.52	100
Prevalence $\pi = 0.10$, $n_P = 25$	0.966	0.287	0.268	+3.87	+2.55	100
Prevalence $\pi = 0.10$, $n_P = 100$	0.976	0.209	0.193	+1.04	-0.57	100
Prevalence $\pi = 0.10$, $n_P = 500$	0.977	0.212	0.196	-0.04	-1.16	100
Prevalence $\pi = 0.50$, $n_P = 25$	0.730	0.185	0.129	+5.33	+4.83	100
Prevalence $\pi = 0.50$, $n_P = 100$	0.810	0.160	0.100	+2.38	+1.59	100
Prevalence $\pi = 0.50$, $n_P = 500$	0.828	0.189	0.115	+0.46	-0.03	100
Curved boundary	0.940	0.167	0.142	-0.94	-2.45	100
Feature interaction Interaction	0.921	0.127	0.110	-9.65	-12.51	100
Unequal class covariance	0.912	0.132	0.116	-4.22	-9.02	100

Note. TN retained / available is both true-negative recall and the fraction of available true negatives retained. U retention includes hidden positives. Means over 100 repetitions. Scenario identifiers correspond to original benchmark. RN masks were reconstructed from exact archived data/code because selected indices were not originally stored.